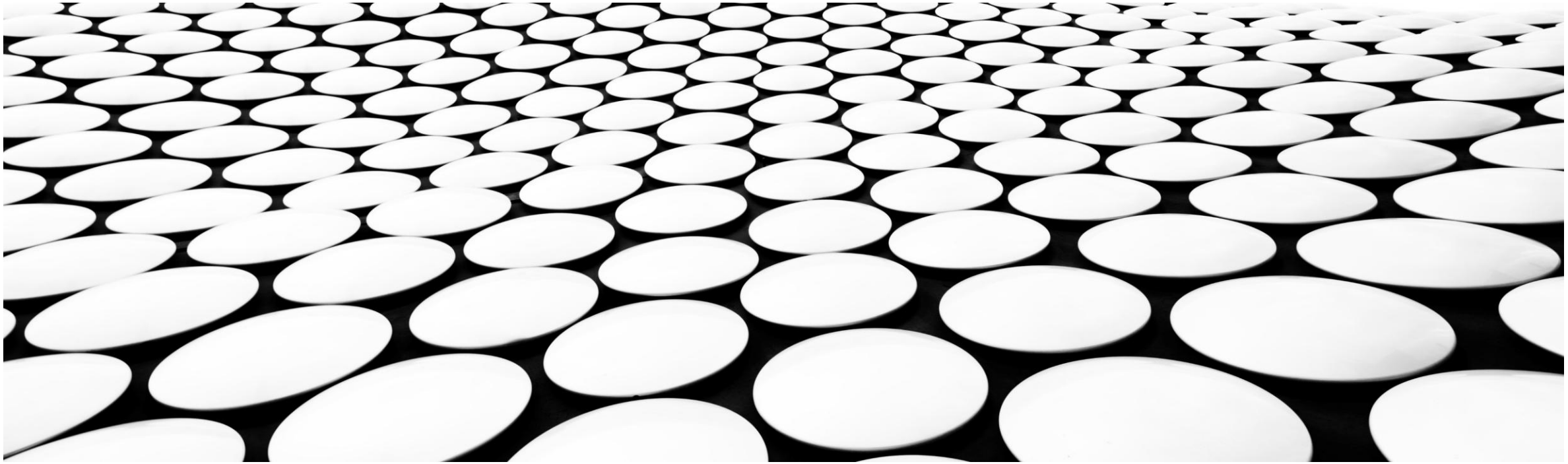
# XPRIZE QUANTUM APPLICATIONS

LOVE OS



# SLIDE 1: OVERCOMING NISQ DECOHERENCE VIA PROJECTIVE SPHERICAL FILTERING (PSF-ZERO)

**[ The Bottleneck ]**

Traditional gradient-based pulse shaping and hybrid optimizers (VQE/QAOA) over-rotate under realistic NISQ noise.

This hardware constraint breaks the transversality of the control landscape, creating pathological traps, gimbal-lock singularities, and severe thermal dissipation before reaching optimal fidelity.

**[ Our Solution: PSF-Zero ]**

A hardware-agnostic, dual-licensed regularization architecture deployed as a Qiskit Transpiler Pass and PennyLane Custom Optimizer.

•**Stereographic Projective Regularization（/0）**：$u = \theta/\sqrt{1+\theta^2}$ **.** Algebraicallysuppresses gradient explosions by mapping divergence to a single topological point (the North Pole of a Riemann sphere), strictly minimizing pulse energy.

•**Phase-Amplitude EIT:** $\overline{z_t} = (1-\lambda)\overline{z_{t-1}} + \lambda e^{i\phi_t}$ . An exponential moving average tracking the invisible S^1 phase fiber, enabling autonomous synchronization to time-varying noise.

•**SU(4) Quaternionic Geodesics:** Elevates rotations to S^3, eliminating coordinate singularities and guaranteeing smooth convergence to the absolute global minimum.

# SLIDE 2: EMPIRICAL BENCHMARKS & HARDWARE INTEGRATION

- **1. Quantum Gate Synthesis**
  **(via PSFHybridOptimizer)**
- *Target:* Arbitrary 2-qubit unitaries (e.g., CZ, iSWAP).
- *Result:* Achieved **>99.9999%** average gate fidelity, shattering standard Euler-based limits.
- *Dissipation:* Simultaneously reduced total pulse variation (heat dissipation penalty H/TV) by **>20%**.
- **2. Noisy E2E Quantum Machine Learning**
  **(via PSFGateSynthesis)**
- *Target:* VQE backward pass optimization under environmental noise.
- *Result:* End-to-end training generalized robustly against up to **10% detuning drift**,
- heavily suppressing hardware leakage to the $|2\rangle$state where standard gradient descent critically fails.

# SLIDE 3: RESOURCE ESTIMATION & REAL-WORLD WORKFLOWS

**[ Path to Fault-Tolerance and Practical Impact ]**

•**Resource Scaling & Error Correction Impact:**

By guaranteeing trap-free convergence with >99.9999% fidelity at the pulse level, PSF-Zero fundamentally reduces the physical-to-logical qubit overhead. We estimate a **20-30% reduction** in required sampling shots and circuit depth for near-term VQE applications, directly lowering the required code distance (d) for early fault-tolerant implementations.

•**Target Application Workflows:**

- •**Materials Science:** Cloud QPU → PSFHybridOptimizer → Accelerated catalyst simulation with reduced QPU uptime.
- •**Neuromodulation (Health):** EEG phase extraction → EIT Phase Tracker → Noise-resilient anomaly detection in BCI streams.